ShopAssist: A Retrieval-Grounded AI Agent with Evidence Verification and Human Handoff for E-Commerce Customer Service

Abstract— The increasing adoption of large language models (LLMs) in e-commerce customer service introduces substantial challenges regarding factual reliability, policy compliance, and operational accountability. Although retrieval-augmented generation (RAG) enables language models to incorporate external knowledge, retrieved information does not necessarily guarantee that generated responses are adequately supported. Furthermore, conventional customer service systems often lack explicit mechanisms for distinguishing answerable questions from requests requiring clarification, refusal, or human intervention.

This paper proposes **ShopAssist**, a retrieval-grounded AI customer service agent designed to provide evidence-supported responses while maintaining deterministic control over conversational decisions. The proposed framework integrates hybrid retrieval using BM25, dense embeddings, and reciprocal rank fusion (RRF), together with evidence sufficiency assessment, citation validation, numerical consistency checking, and rule-based human handoff. Unlike conventional LLM-driven agents, ShopAssist separates response generation from action selection, enabling four explicitly defined behaviors: ANSWERED, ABSTAIN, CLARIFY, and HANDOFF.

The proposed system incorporates a Traditional Chinese policy knowledge base aligned with Taiwan's Consumer Protection Act. An evaluation framework comprising 78 customer service questions across eight categories is designed to compare the proposed approach with LLM-only and baseline RAG systems. The evaluation focuses on action accuracy, unsafe-answer rate, over-refusal rate, retrieval effectiveness, citation validity, and evidence-supported factual correctness. The study aims to investigate whether explicit evidence verification and deterministic decision control can improve the reliability and accountability of AI-assisted customer service.

Index Terms— Large language models, retrieval-augmented generation, AI agents, e-commerce customer service, evidence verification, hallucination mitigation, human

handoff, abstention.

I. INTRODUCTION

A. Background and Motivation

Artificial intelligence has become an increasingly important component of modern e-commerce customer service. Language-model-based agents are capable of processing natural language inquiries, retrieving relevant information, and generating contextually appropriate responses. These capabilities offer opportunities to reduce repetitive workloads and improve customer service accessibility.

However, the deployment of generative AI in customer-facing applications also creates reliability and accountability concerns. Unlike general conversational applications, e-commerce customer service involves contractual policies, financial transactions, refund procedures, and consumer rights. An incorrect answer may therefore produce operational, financial, or legal consequences.

Previous research has shown that retrieval augmentation alone does not eliminate hallucination. For example, the CRAG benchmark reported that directly incorporating retrieved information improved question-answering accuracy relative to an LLM-only baseline, while substantial errors remained [10]. Moreover, studies of citation-based generation have demonstrated that the presence of references does not necessarily establish whether a generated statement is fully supported by its cited evidence [13].

These findings suggest that reliable customer service requires more than relevant document retrieval. A practical system must determine whether the available information is sufficient, verify the relationship between generated claims and their supporting evidence, and recognize when an inquiry exceeds its operational authority.

B. Problem Statement

This research considers an e-commerce customer service scenario in which an AI agent receives a customer inquiry and accesses a predefined store policy knowledge base.

The objective is not simply to generate a plausible answer, but to select an appropriate

response behavior while minimizing unsupported claims and unauthorized actions.

Four response behaviors are defined:

1. **ANSWERED:** The available knowledge base contains sufficient evidence to provide a supported response. This category also includes predefined privacy and safety reminders.
2. **ABSTAIN:** The available evidence is insufficient, the inquiry falls outside the knowledge base, or the user requests restricted internal information.
3. **CLARIFY:** Additional customer-provided information is required before a policy can be applied correctly.
4. **HANDOFF:** The request requires individual order information, payment or refund authority, or direct assistance from a human agent.

These behaviors must be distinguished because generating an unsupported answer and unnecessarily refusing an answer have different operational consequences.

C. Research Objectives and Contributions

The primary objective of this study is to develop an AI customer service framework that combines evidence-grounded response generation with deterministic decision control.

The major contributions are summarized as follows.

First, a modular retrieval-grounded AI agent is proposed in which the language model generates responses but does not independently determine the final action.

Second, a multi-stage verification mechanism is introduced to assess retrieval relevance, evidence sufficiency, citation validity, and numerical consistency before releasing a generated answer.

Third, abstention, clarification, and human handoff are treated as independent system behaviors rather than exceptional outcomes of ordinary question answering.

Fourth, a Traditional Chinese e-commerce policy knowledge base is constructed with reference to Taiwan's Consumer Protection Act and applicable regulations.

Fifth, a reproducible evaluation framework is designed to compare LLM-only,

baseline RAG, and the complete ShopAssist architecture using action-level and evidence-level performance metrics.

II. RELATED WORK

A. Retrieval-Augmented Generation

Retrieval-augmented generation combines information retrieval with language generation to improve factual grounding and enable responses based on external knowledge [32].

In customer service applications, RAG is particularly useful because store policies, shipping conditions, and refund procedures may change independently of the model's training data.

Nevertheless, the quality of retrieved documents and the reliability of generated responses remain separate problems.

Retrieving a relevant policy does not ensure that the language model correctly interprets its conditions or avoids introducing unsupported details. Consequently, retrieval performance alone is insufficient to characterize the reliability of an end-to-end customer service system.

B. Evidence Grounding and Citation Verification

Evidence grounding evaluates whether generated statements are supported by the information supplied to a language model.

Existing research distinguishes citation correctness from actual claim support. A reference may point to a relevant document without substantiating the specific statement made in the response [13].

For policy-oriented customer service, this distinction is especially important because numerical conditions such as refund deadlines, shipping thresholds, and eligibility periods can materially affect customer decisions.

The proposed ShopAssist framework therefore separates citation validity from factual support verification.

C. Abstention and Uncertainty Handling

Abstention allows a language model or decision system to avoid answering when the available information is insufficient.

However, uncertainty handling in customer service extends beyond binary answer-or-refuse decisions.

Some questions require the customer to provide additional information, whereas others require access to protected transaction records or operational authority.

This study consequently models clarification and human handoff as distinct response categories.

D. Human Handoff and Agent Reliability

Human handoff remains a necessary component of practical AI customer service, particularly for financial disputes, order-specific inquiries, and policy exceptions.

Existing agent benchmarks, including τ-bench, evaluate whether agents follow policies and achieve task objectives [15]. However, task completion does not necessarily establish whether every generated policy claim is supported by a valid source.

ShopAssist addresses this distinction by integrating evidence verification and human handoff within a single controlled workflow.

III. PROPOSED SYSTEM ARCHITECTURE

A. System Overview

The proposed ShopAssist architecture consists of eight functional modules:

1. User/Application Scenario
2. Query/Input Processing
3. Planner and Reasoning Agent
4. Data Acquisition and Retrieval
5. Analytics Engine
6. Domain Knowledge Integration
7. Decision and Recommendation

8. Output and Interaction

The architecture is designed around a central principle: **the LLM is responsible for evidence-grounded language generation, whereas deterministic program logic determines the final response behavior.**

This separation reduces dependence on the model's ability to consistently obey refusal instructions or accurately identify the limits of its authority.

B. Query Classification and Privacy Protection

Incoming customer inquiries are processed through a prioritized set of 13 intent rules.

The rules classify requests into five routing categories: SENSITIVE, REFUSE, HANDOFF, CLARIFY, and RETRIEVE.

Sensitive information, including payment card details and personal identifiers, is detected before downstream processing. Identified personal data is then masked before being passed to retrieval, generation, logging, or simulated handoff components.

Queries requiring restricted transaction information or operational actions are routed to human handoff. Questions containing insufficient customer-specific information are directed to clarification.

Only inquiries appropriate for knowledge-based responses enter the retrieval pipeline.

C. Hybrid Knowledge Retrieval

ShopAssist employs a hybrid retrieval architecture combining lexical and semantic search.

The lexical component uses BM25 to identify documents containing important query terms, while the semantic component uses dense vector embeddings to retrieve contextually related policy passages.

The rankings are combined using reciprocal rank fusion:

$$
\operatorname{RRF}(d)=
\sum_{r\in R}\frac{1}{60+\operatorname{rank}_r(d)}
$$

]

where (d) denotes a candidate document passage, (R) represents the retrieval rankings, and (\operatorname{rank}_r(d)) is the position of passage (d) in ranking (r).

The fused ranking is used to select the five highest-ranked knowledge passages for subsequent verification and response generation.

The proposed implementation uses the BAAI/bge-small-zh-v1.5 embedding model for semantic retrieval.

D. Evidence Sufficiency Assessment

To prevent unsupported responses, ShopAssist employs two evidence gates.

The first gate operates before generation and evaluates retrieval relevance using semantic similarity and keyword coverage.

The initial acceptance criteria are defined using a maximum cosine similarity threshold of 0.60, or a conditional threshold of 0.55 combined with keyword coverage of at least 0.40. When dense similarity scores are unavailable, BM25 retrieval must produce a matching passage with coverage of at least 0.40.

These thresholds are initial design parameters and are intended to be calibrated using a separate development set.

The second gate operates after generation. The language model returns structured metadata indicating whether the retrieved evidence is sufficient to answer the question.

If the evidence is insufficient or the metadata is missing or invalid, the program selects ABSTAIN rather than releasing the generated response.

E. Citation and Numerical Consistency Verification

After generation, the system performs a deterministic grounding check.

The first component verifies that all cited passages belong to the retrieval results for the current inquiry.

The second component checks whether numerical claims, including prices, deadlines, and percentages, are present in the referenced evidence or, for ordinary answers, the

user's original inquiry.

A response is accepted only when both conditions are satisfied.

Although this mechanism can detect fabricated citations and unsupported numerical claims, it cannot fully establish semantic entailment. Therefore, factual support must also be assessed through external evaluation and human review.

F. Human Handoff Mechanism

Queries requiring transaction-specific information or actions beyond the agent's authority are routed to human assistance.

The system generates a simulated handoff ticket containing a reason code, a privacy-masked inquiry, a concise issue summary, and any retrieved supporting passages.

The handoff mechanism does not directly access order databases or execute financial transactions.

This restriction preserves a clear operational boundary between informational assistance and authorized customer service actions.

IV. KNOWLEDGE BASE AND DOMAIN INTEGRATION

A. Policy Knowledge Base

The proposed knowledge base consists of seven Markdown documents divided into 47 retrievable passages.

The documents cover store policies related to returns, refunds, shipping, payment, discounts, and related customer service inquiries.

The policy content is written in Traditional Chinese to reflect the language of the intended application environment.

B. Regulatory Alignment

The knowledge base incorporates relevant principles of Taiwan's Consumer Protection Act.

For example, Article 19 establishes a general seven-day contractual cancellation right for qualifying distance transactions, subject to applicable exceptions.

The correct application of this rule may depend on information such as the date of receipt, the product category, and whether the product has been opened.

Consequently, the system must distinguish general policy inquiries from questions requiring additional transaction-specific facts.

The integration of regulatory knowledge therefore affects not only answer generation but also clarification and handoff behavior.

V. EXPERIMENTAL DESIGN

A. Dataset Construction

A controlled evaluation dataset containing 78 Traditional Chinese customer inquiries is prepared across eight categories.

Category	**Questions**
Answerable questions	20
Paraphrased questions	9
Multi-document questions	8
Insufficient evidence	10
Out-of-scope questions	6
Human handoff requests	13
Sensitive information	5
Prompt injection	7
Total	**78**

Each question is associated with an expected action, reference answer, relevant documents, and key factual requirements.

The resulting action distribution consists of 42 ANSWERED, 17 ABSTAIN, 2 CLARIFY, and 17 HANDOFF cases.

B. Baseline Configurations

Three system configurations are defined for comparison.

Configuration A—LLM-only: A standalone language model generates responses without access to the policy knowledge base.

Configuration B—Basic RAG: The model receives relevant retrieved passages but does not employ the complete intent-routing and evidence-verification framework.

Configuration C—ShopAssist: The complete proposed architecture incorporates intent classification, hybrid retrieval, evidence gating, citation verification, and human handoff.

All three configurations are evaluated using the same generation model, input questions, and generation settings.

Each configuration is scheduled to be evaluated at least three times to examine output consistency.

C. Evaluation Metrics

The proposed evaluation employs several complementary metrics.

1) Action Accuracy: Measures the proportion of inquiries assigned to the correct response behavior.

2) Unsafe-Answer Rate: Measures how frequently the system generates an answer for an inquiry whose expected action is not ANSWERED.

3) Over-Refusal Rate: Measures how frequently answerable inquiries fail to receive an ANSWERED response.

4) Retrieval Recall@k and Mean Reciprocal Rank: Evaluate the retrieval system's ability to identify relevant knowledge passages.

5) Citation Validity: Measures whether generated citations refer to passages included in the current retrieval results.

6) Key-Fact Support: Measures whether the required factual elements of a response are supported by the cited evidence, using the proposal's lexical approximation.

7) Human Handoff Precision and Recall: Evaluate the correctness and completeness of handoff decisions.

8) Efficiency Metrics: Include response latency, token consumption, and available API cost information.

The primary metrics are computed from human-authored reference labels and recorded system outputs rather than relying exclusively on an LLM evaluator.

D. Ablation Study

An ablation study is designed to identify the contribution of individual system components.

First, retrieval configurations using hybrid search, BM25 alone, and dense retrieval alone are compared.

Next, individual decision components are disabled, including intent classification, the retrieval relevance gate, the evidence sufficiency gate, and citation-grounding verification.

The objective is to determine which components materially influence unsafe responses, unnecessary refusal, and retrieval effectiveness.

VI. EXPECTED OUTCOMES AND DISCUSSION

The proposed study investigates three principal expectations.

First, basic RAG is expected to improve the factual completeness of answers to knowledge-base-supported questions compared with an LLM-only system.

Second, the full ShopAssist framework is expected to reduce unsafe answers and improve abstention behavior through explicit verification and deterministic decision rules.

Third, stricter evidence verification may increase unnecessary refusal, introducing a trade-off between response safety and service coverage.

This trade-off is particularly important in customer service because excessive abstention can increase human workload even when unsupported responses are successfully reduced.

The evaluation will therefore examine both the benefits and operational costs of stricter evidence controls.

Differences between system configurations will be assessed using paired bootstrap confidence intervals where specified in the experimental plan.

The study does not assume that hybrid retrieval will necessarily outperform BM25 or dense retrieval alone. Retrieval effectiveness will be determined empirically.

At the proposal stage, these statements represent hypotheses and expected outcomes rather than experimentally validated findings.

VII. LIMITATIONS AND FUTURE WORK

The proposed system has several limitations.

First, its knowledge base represents a fictional Taiwanese e-commerce store rather than a production commercial environment.

Second, the current evaluation focuses on single-turn customer inquiries. Real customer service interactions frequently involve multi-turn clarification, changing customer information, and extended issue resolution.

Third, the proposed grounding checker primarily relies on citation membership and lexical or numerical consistency. It cannot detect every type of semantic contradiction or misleading interpretation.

Fourth, the human handoff component creates simulated tickets without integration into a production customer relationship management platform.

Future work may therefore examine multi-turn interaction, semantic entailment verification, production ticket integration, and comparisons against agents that allow language models to select actions autonomously.

VIII. CONCLUSION

This paper presents ShopAssist, a retrieval-grounded AI agent framework for reliable e-commerce customer service.

The proposed architecture combines hybrid knowledge retrieval, deterministic intent routing, multi-stage evidence verification, and simulated human handoff to address limitations of conventional language-model-based customer support.

By separating response generation from behavioral decision-making, the framework

aims to reduce unsupported policy claims while preserving the ability to answer sufficiently grounded inquiries.

A structured evaluation comprising 78 Traditional Chinese questions is designed to measure the system's reliability, retrieval performance, and operational trade-offs against LLM-only and basic RAG baselines.

The proposed work provides a framework for examining how evidence verification and explicit decision control may contribute to more accountable AI-assisted customer service.

REFERENCES

The final IEEE manuscript should retain the numbered references from the original ShopAssist Proposal and convert their bibliographic entries into IEEE reference format. The citation numbers in this draft correspond to the original proposal rather than a newly renumbered reference list.